

International Soccer Team Performance Analytics

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Executive Summary

Soccer is one of the biggest sports in today's society. It can be difficult to keep track of the analytics, which makes it a challenge for analysts to make well-informed decisions that can influence team dynamics and performance. The different patterns related to wins, losses and draws, type of tournament and even time periods can be analyzed to aid those decisions. This project focuses on developing a structured data-driven approach to analyze soccer teams' and match data using dimensional modeling and visual storytelling.

A centralized data model was created using match-level data from the International Football Results dataset on Kaggle and doing data preparation using Python. We standardized team and location names, aligned datatypes and formats, filtered to a set time period, 2011-2025, the most recent 15 years of data then removed duplicate and missing data. A dimensional model was constructed following Kimball methodology centered around match results with dimensions for team, date, tournament and location. Dimensional modeling was chosen over a transactional model for efficiency in supporting analytical queries and trend analysis.

This analysis uncovered crucial insights. First, we covered team performances, revealing that over time, performances varies. There are stronger teams like Argentina who demonstrates consistent wins, while others do not. We conducted a tournament-based analysis which highlighted that performance varies based on the stakes involved. We noticed when the stakes are high, teams with historical high performance tend to outperform other teams with similar history but with low goal differences which makes the match so interesting and classified as "high-stakes". The analysis also brought out the idea of a home field advantage as we observed most teams having the majority of their wins on their home field rather than away. Lastly, when we looked at the goal differences, it showed that most games were decided by small margins, which means that any adjustments in strategy could have significant impacts on results. This is a unique window of opportunity for decision makers.

These findings support the project problem statement by showing that improving decision making in soccer is needed. It can be done by using data modeling with visualization to provide a comprehensive and innate view of soccer team performance. Coaches, analysts other stakeholders can use these insights to evaluate their team prepare for future games to achieve the desired outcomes.

All in all, this shows how unified analytical framework can transform soccer data into actionable insights. This could be done by combining SQL-based data modeling with interactive visual storytelling which would enable informed data driven that can enhance team performance and a competitive strategy.

Problem Statement

Team Performance Optimization

Soccer teams across the world compete in various tournaments and league games, generating loads of data over time. However, analyzing data consistently across time, tournaments, match conditions, scoring dynamics, and opponents is becoming increasingly difficult without a unified framework to evaluate the performance of the teams. Data analysts and scientists tend to struggle to identify clear patterns in wins, losses, and draws.

For these soccer teams, performance trends in their historical data have a direct impact on team rankings and decisions with regards to planning and overall success, so it is crucial to address this analytical challenge in order to aid the decision-making process when it comes to team performance and strategy.

Our goal with this project is to use a dimensional data model with statistical modeling and storytelling to analyze match results across teams, time, tournaments, match conditions, and other factors to uncover meaningful insights and trends and to support better, data-driven decision-making.

Literature Review

Based on research, the use of data analytics in soccer has become very popular over the past few years. It is very difficult to collect data based on team performances across all competitions, and time because of a lack of standardized, structured data frameworks across leagues/tournaments. Businesses that use data analytics make better decisions than businesses that don't, according to Davenport and Harris (2019). This basically brought about the idea of sports teams trying to do better by looking at the performance insights of other teams.

In the field of soccer analytics, it was shown that when evaluating performance, you should use a combination of variables, like match results, opponent strength, and environmental factors. There are also contextual factors like the location and the standard of the opposition, which usually have a big impact on team performance (Largo, 2010). It can also be seen how data-driven methods can reveal soccer's hidden patterns, like home and away advantages as well as technical efficiencies that are difficult to see through conventional analysis (Sally 2013).

From a data management standpoint, dimensional modeling offers a strong framework for structuring big data sets to facilitate analytical queries. Fact and dimensional tables make up schema designs that enable effective querying and trend analysis (Kimball, 2013). For sports analysis, this is quite useful because it captures data from a variety of different angles, including time, teams, and tournaments. Performance trends can be spotted by analysts when they use dimensional models.

For the purposes of this project, we made use of Python, Oracle SQL Developer, and Power BI to build a data warehousing framework. This approach highlighted the way that dimensional modeling and data visualization go hand in hand to convert a complex dataset into valuable insights and provide team leadership with the necessary data insights to drive strategy and performance.

In new sports studies, analytics highlights the increased significance of performance and predictive insights. Big data analytics can change how decisions are made in a variety of industries. Organizations can find patterns and create recommendations that would improve competition by using historical match data in a framework (Goes, 2014).

Data Collection and Preparation

The dataset used in this project is the International Football Results dataset uploaded to Kaggle by Mart Jürisoo. The dataset contained records in a couple of tables from international soccer matches spanning the period 1872 – 2026. This data covers everything from match dates, teams, scores, tournament names to the location of the matches, and historical team name mappings which help with context. This dataset was chosen due to its coverage of international data and its suitability to our goals.

We performed our data preparation using Python to clean the raw data and transform it for analysis. Here we focused on ensuring consistency and alignment throughout the data. Firstly, we loaded the tables from Kaggle and performed checks to examine the overall structure of the data, especially the datatypes, so we could make transformations where necessary.

Following that, the first step we took to prepare the data was to standardize the datetime fields and ensure they were all the same datatype for analysis. This step was crucial for our upcoming work. We then created mappings to ensure consistency in the names of the teams and location names.

The next vital step was for us to address the time period we would analyze so we went ahead to filter the dataset to only include 15 of the most recent years of data from 2011 to 2025. At this point, the `former_names` table revealed that two teams had name changes during the time period we selected, so we took steps to account for those changes in the aforementioned team name mapping.

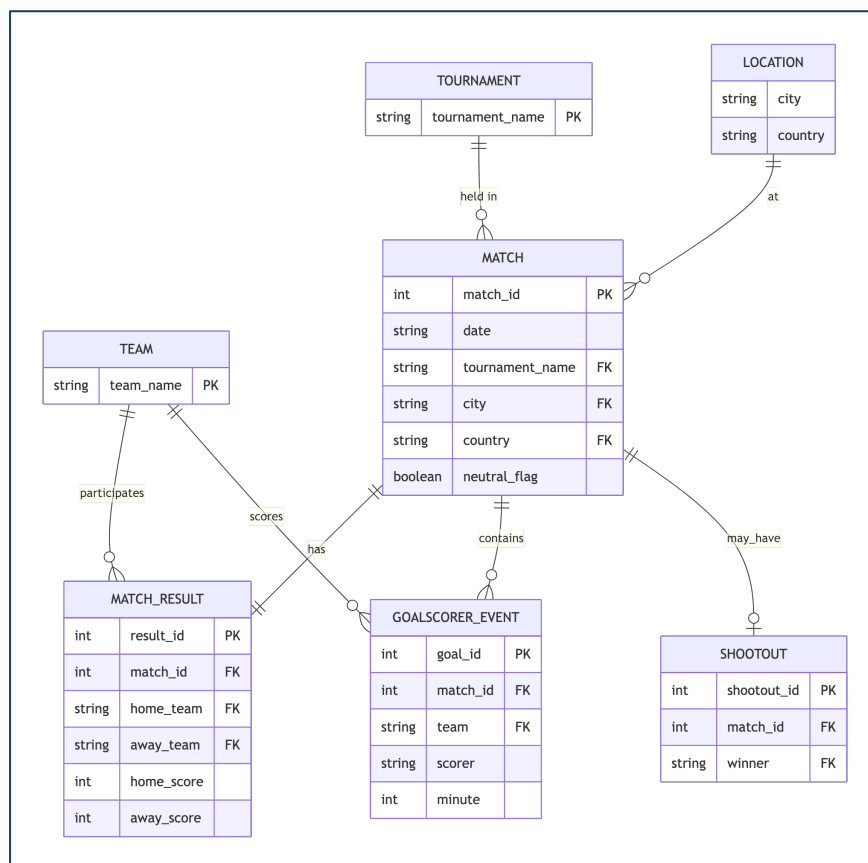
We went on to check the data for missing values and decided to remove the small number of rows we found at that point since we determined they would not have any significant impact on our analysis. We then deduped the data and proceeded with validation and summary checks including shape, size, types and counts for the dataset. The cleaned data was saved in csv files for use in our analysis.

Database Design

This database design properly supports the analytical goal of evaluating the performance of international soccer teams across tournaments, time, and other factors. To meet the requirements of this project, we will build transactional and dimensional models. The transactional model shows just how detailed operational data can be organized to record event data for soccer games. The purpose of this model is to show how detailed records can be stored and organized. On the other hand, the dimensional model supports the same data, but in a Kimball-style, analytical structure better suited for trend analysis and decision support.

Transactional Model

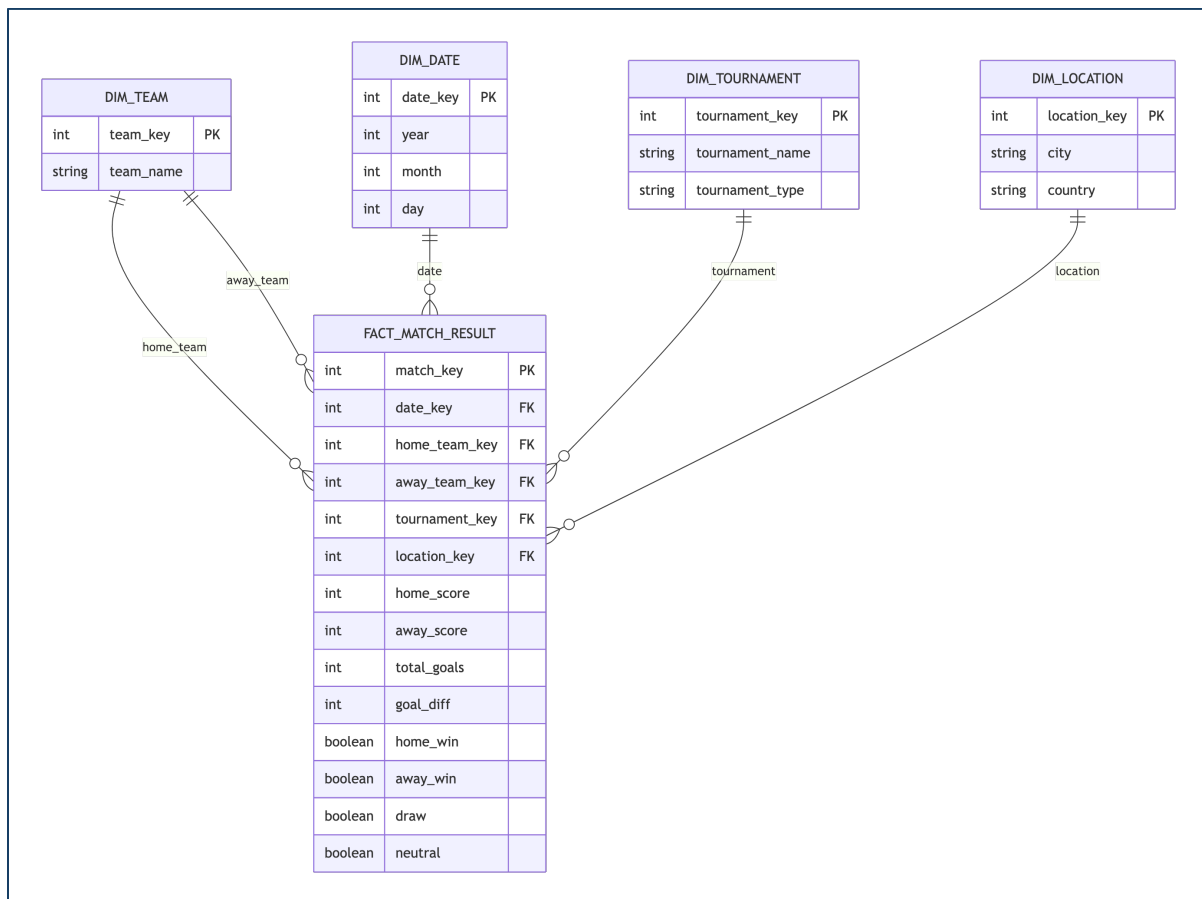
This transactional model is designed to keep detailed records in a normalized structure. The model captures matches, tournaments, teams, goal score events, and penalty shootouts are maintained separately. Here, we have the match table representing the core event. The supporting tables hold additional details such as the team information, tournament data, location data, match results, and penalty shootout events. A structure like this reduces redundancy and is suitable for a system that records match-level data over time.



Although the transactional model is the best for recording details, it's not necessarily the best for an analytical project like this one. The goal of this project's work is to analyze identify trends in team performance across multiple dimensions. As much as the transactional model is great for comparison, as we move to answering analytical questions to achieve the goals of the project, the transactional design is less intuitive and less efficient for reporting the insights needed.

Dimensional Model

The dimensional model for this analysis is specifically designed to support analytical querying, aggregation, and storytelling. A well-organized Kimball style star schema will help us to identify the trends in performance we are searching for. The schema is constructed around the fact_match_result table which stores measurable outcomes of each match and has a grain of one row per international soccer match. This grain allows us to have a row representing a single match between two teams, on a specific date, at a specific tournament and at a specific location, hence creating/defining the dimensions for the model.

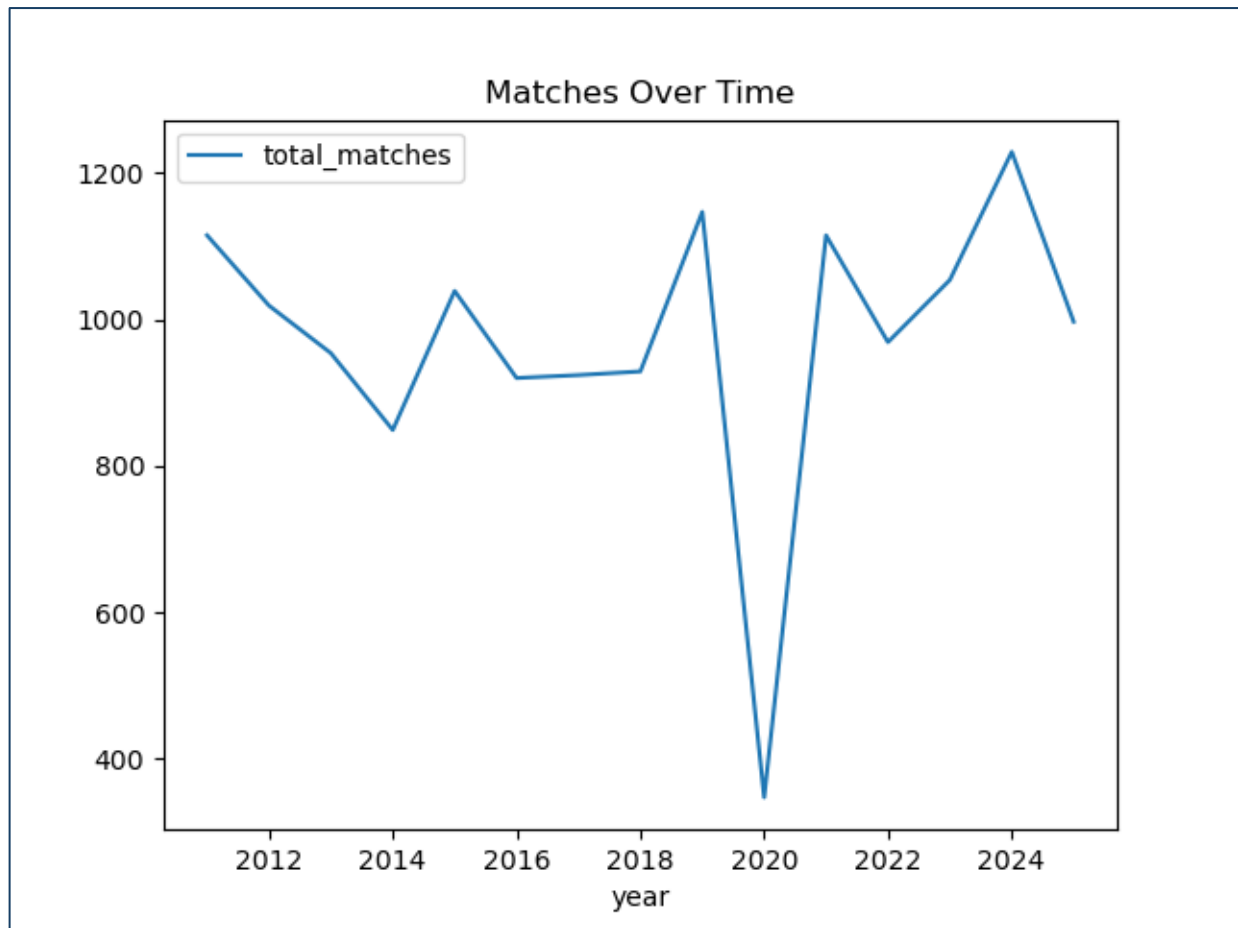


Overall, the differences in the transactional and dimensional models have shown that transactional models are quite suitable for detailed record keeping when the dimensional model is ideal for analytical purposes like answering managerial questions efficiently.

Exploratory Data Analysis

Some brief Exploratory Data Analysis was conducted using Python to better understand the structure, distribution, and patterns of the data we were working with.

```
=== Matches Over Time ===  
   year  total_matches  
0  2011           1115  
1  2012           1019  
2  2013            954  
3  2014            849  
4  2015           1039  
5  2016            920  
6  2017            924  
7  2018            929  
8  2019           1147  
9  2020            347  
10 2021           1115  
11 2022            969  
12 2023           1054  
13 2024           1229  
14 2025            997
```



=== Match Outcomes ===

```
home_win    7000
away_win    4231
draw        3376
```

=== Goal Statistics ===

```
total_goals  2.730061
goal_diff    1.652222
```

=== Top Teams by Matches Played ===

	team	matches_played
175	Mexico	255.0
288	United States	236.0
215	Qatar	233.0
63	Costa Rica	218.0
138	Jordan	218.0
128	Iraq	217.0
204	Panama	216.0
234	Saudi Arabia	208.0
199	Oman	208.0
21	Bahrain	207.0

=== Home Win Rate by Team ===

	home_team	home_games	home_wins	win_rate
78	Egypt	105	83	0.790476
252	Spain	100	75	0.750000
135	Jersey	21	15	0.714286
26	Belgium	97	68	0.701031
93	France	115	80	0.695652
11	Argentina	117	81	0.692308
178	Morocco	119	82	0.689076
125	Iran	96	66	0.687500
36	Brazil	115	79	0.686957
4	Algeria	98	67	0.683673

=== Matches by Tournament ===

	tournament	total_matches
0	Friendly	4445
1	FIFA World Cup qualification	3406
2	African Cup of Nations qualification	1002
3	UEFA Euro qualification	922
4	UEFA Nations League	658
5	CONCACAF Nations League	422
6	African Cup of Nations	320
7	AFC Asian Cup qualification	250
8	Gold Cup	225
9	COSAFA Cup	201

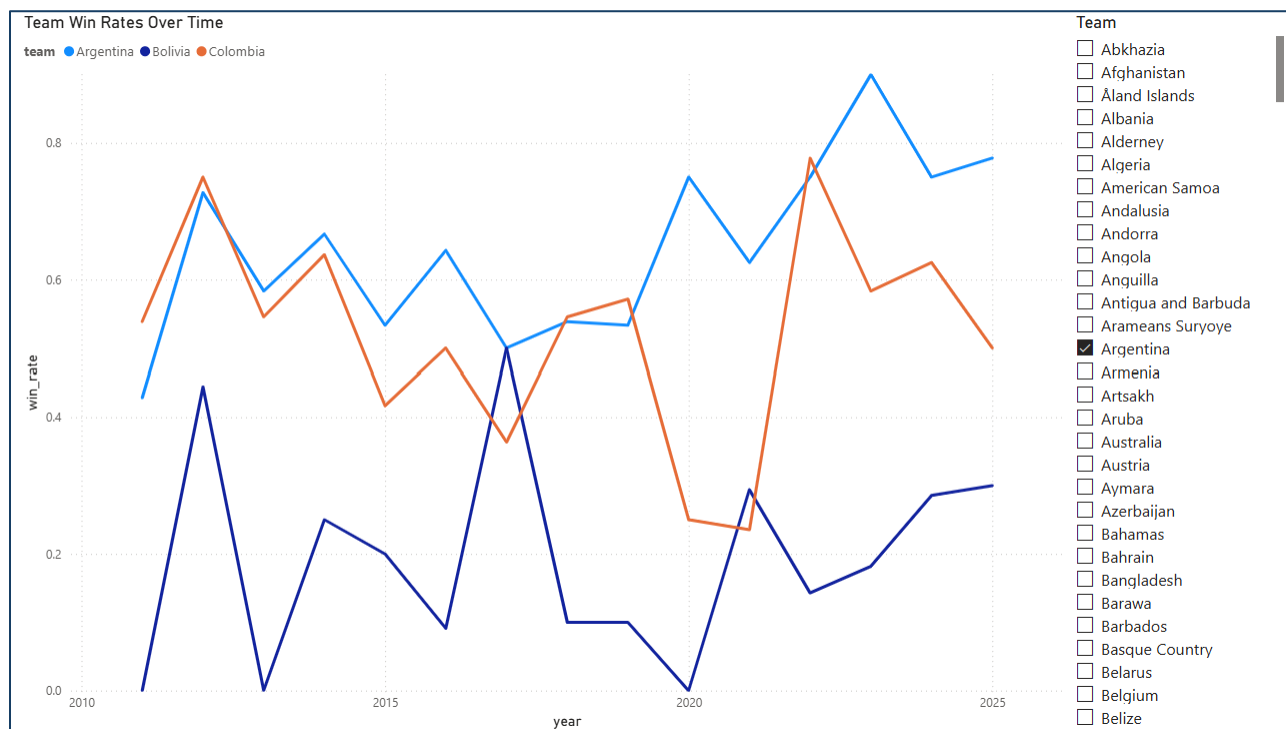
=== Top Countries by Matches ===

	country	total_matches
0	United States	661
1	Qatar	463
2	United Kingdom	463
3	South Africa	339
4	United Arab Emirates	304
5	Morocco	283
6	France	234
7	Brazil	180
8	Russia	165
9	Germany	160

Reporting & Storytelling

We've worked on putting together a data-driven narrative for international soccer team performance. Our analysis focuses on identifying patterns across time, tournaments, match conditions, and scoring dynamics. The following visualizations illustrate key insights that support the decision-making process when it comes to team performance.

This visualization showing Team Win Rates over time examines how team performance evolves over time, allowing teams to assess consistency and look at performance trends across multiple teams. This visual directly supports our problem statement's goal of identifying patterns in performance across seasons.

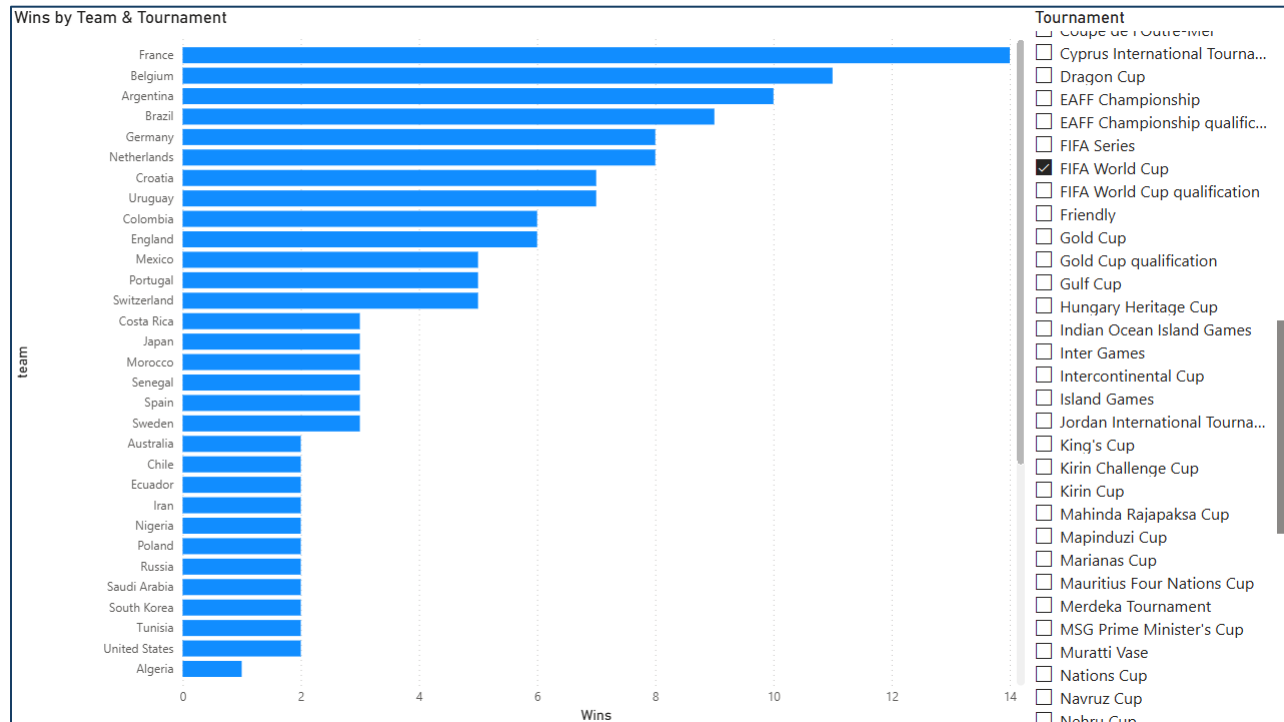


Example insights to be drawn from the figure above include:

- Argentina has shown a consistently strong win rate that has been improving over time, peaking at 90% in recent years.
- Bolivia exhibits low but also inconsistent performance with respect to win rates, frequently landing themselves below 30%.
- Although they have demonstrated moderate win rates over time, Colombia's performance has fluctuated greatly over the 15-year time span.

This analysis highlights which teams (like Argentina for example) are most likely benefitting from stable strategies and player development over time.

The second visualization shows the Number of Wins by Team and Tournament to compare team performance within a specific tournament. This highlights how teams perform under similar conditions. This is useful for decision makers to gather information about their own team but also about their opponents because tournament-specific success often reflects a team's ability to perform in high stakes situations.

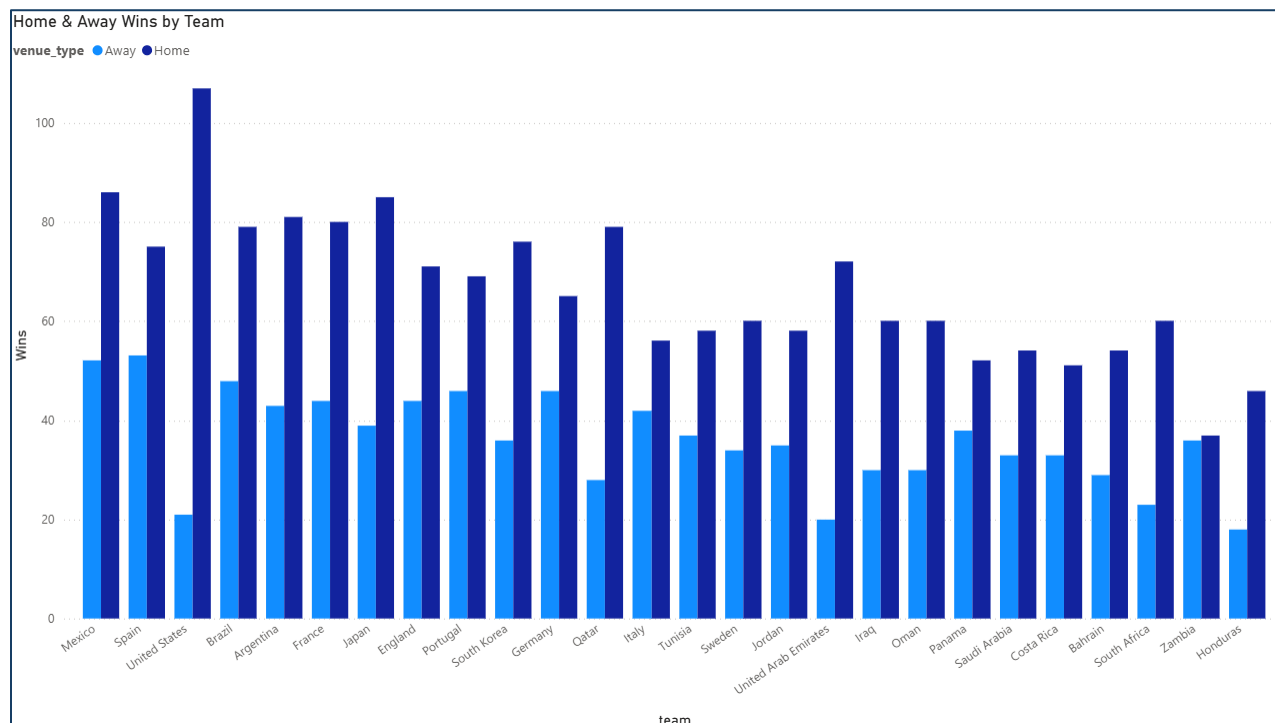


Example insights to be drawn from the figure above include:

- In the past 15 years, within the FIFA World tournament, France has recorded 14 match wins, followed by Belgium with 11 and Argentina with 10 match wins.
- After the top teams, we see a bit of a hierarchal pattern in the bars which can be an indicator of how far into the tournament teams tend to make it.
 - This aligns with the idea that historical tournament performance can guide strategic planning and benchmarking.

Performance tends to vary greatly across tournaments, and a chart like this can be used to assess how teams perform in different tournaments and even in “friendlies” where the stakes are not high providing good context to their performance analysis. Insights like this are quite valuable for an evaluation of competitive strength.

This clustered column chart displays the Number of Home and Away Wins for the soccer teams and highlights the significance of match location, which was a goal of our problem statement. Insights from this chart help to identify the teams who have relatively strong ‘away performance’ as they tend to be more resilient and adaptable. These insights can be used to influence the manner in which teams prepare a schedule packed with travel, etc.

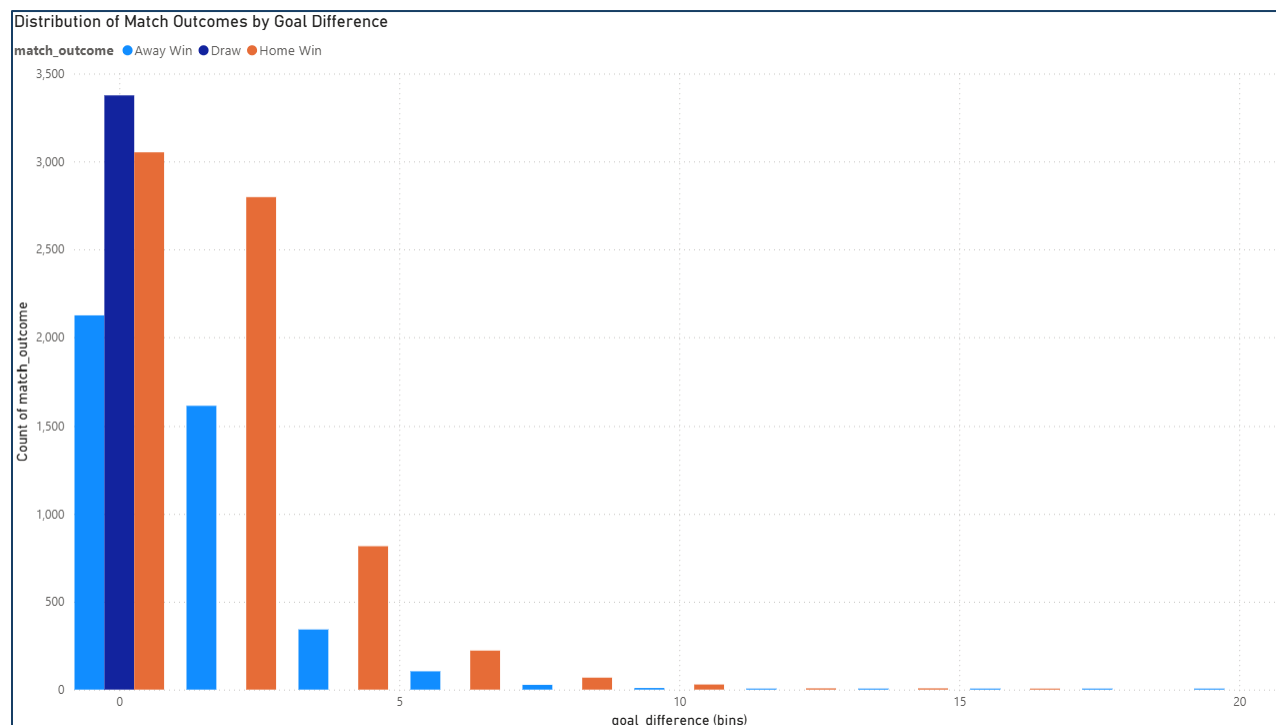


Example insights to be drawn from the figure above include:

- Across the board, the teams display a higher number of home wins than away wins which confirms the presence of a home field advantage in international soccer.
- From the countries presented, the United States shows the largest number of home wins and the largest difference between home and away wins which can signify a high reliance on home field advantage.

The chart shows that home field advantage appears to play a significant role in match outcomes. This could be due to several different reasons ranging from crowd support to reduced travel fatigue and even familiarity or lack thereof, with playing conditions.

This fourth visualization brings context to the margins by which teams win games. It helps to provide insight into the relationship between scoring margin and the result type (Home win, Draw, Away Win). The graph displays how decisive games compare to closely contested ones so we can gain an understanding of how goal margins are related to the match result.

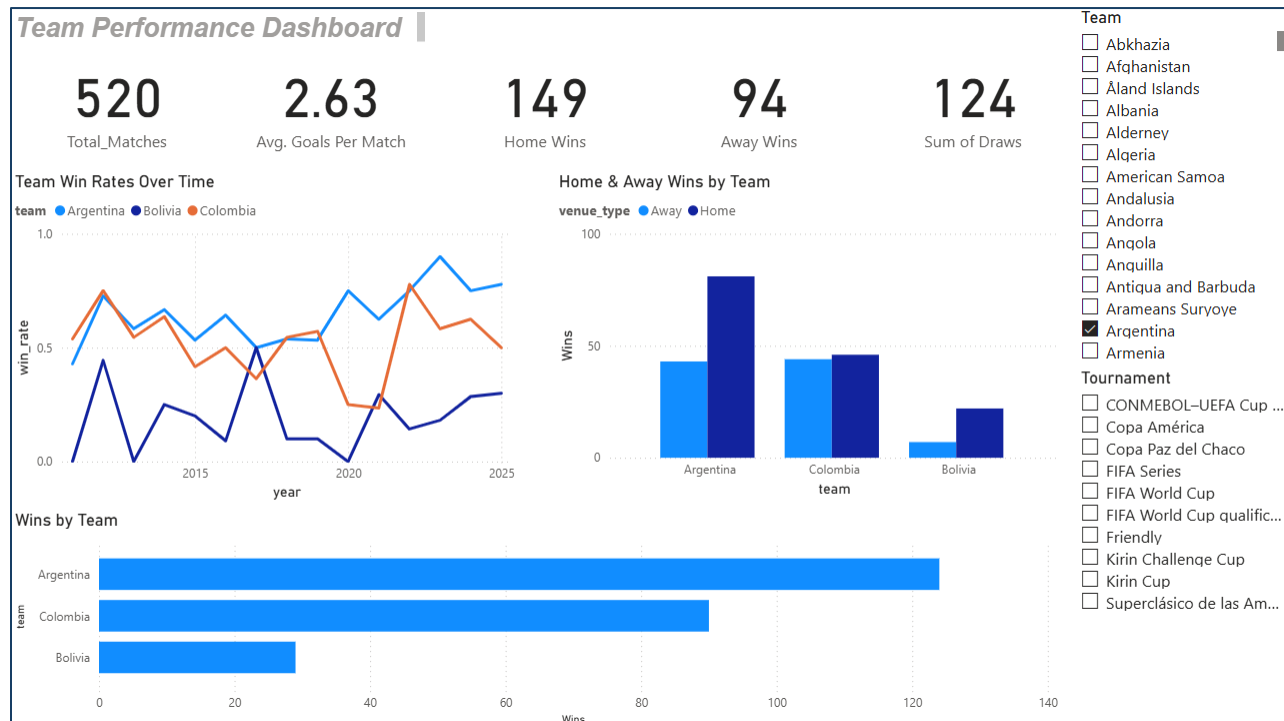


Example insights to be drawn from the figure above include:

- Many of the match outcomes are clustered at low goal differences.
- At higher goal differences, home wins tend to be more frequent than away wins, again communicating the presence of home field advantages.
- Draws obviously occur only at zero goal difference by definition.

Goal difference is an indicator of match dominance that can be further assessed as a part of team performance evaluation.

We've put our top three visuals together with some other key performance indicators to construct a Team Performance Dashboard. This integrates crucial metrics into one interactive interface that can be used to assess performance over multiple dimensions. The dashboard demonstrates the power of enabling flexible analysis by combining multiple perspectives to provide a comprehensive view of team performance.



******This screenshot was taken with Argentina, Bolivia and Colombia selected on the Team slicer.***

Example insights to be drawn from the figure above include:

- Argentina shows the highest total wins among the selected teams
- Collectively, the three teams have 149 Home wins and 96 away wins.
- Over the span of 520 matches, the average number of goals per match was 2.63.

Overall, the insights from this dashboard provide a snapshot of team behavior and can be filtered to include either a single team or a group of teams within or outside of specific tournaments. This can be used to support decision-making in a number of ways to help teams work on their own strategy and also to become more and more familiar with their opponents.

Conclusion

Findings

- Teams exhibit varying performance trends over time with high performing teams tending to maintain consistently high winning rates.
- A strong home advantage exists in the data, with the teams consistently winning more matches on their home field rather than away.
- Performance differs significantly across competitions, signifying the importance of keeping tournament in mind for performance context.

Limitations

- We did not include a match ID field, but it would have been helpful to provide more detailed insights for our analysis.
- External data such as weather and player injuries were not included but could provide additional context in sports analytics.

Future Scope

- Apply predictive modeling to forecast match outcomes based on set conditions. This would allow teams to plan even more strategically based on their opponent and location, etc.
- Incorporate player-level data for individualized performance analysis, which has the potential to assist decision makers with lineups for specific matches.
- Integrate match IDs to allow for further depth in the goal difference analysis.
- Incorporate data on external conditions known to affect sports like weather conditions, etc.

References

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Appendix

SQL Queries

Yearly win rates by team

```
WITH team_match_results AS (
    SELECT
        d.year,
        ht.team_name AS team,
        CASE
            WHEN f.home_win = 1 THEN 1
            ELSE 0
        END AS win_flag
    FROM fact_match_result f
    JOIN dim_date d
        ON f.date_key = d.date_key
    JOIN dim_team ht
        ON f.home_team_key = ht.team_key

    UNION ALL

    SELECT
        d.year,
        at.team_name AS team,
        CASE
            WHEN f.away_win = 1 THEN 1
            ELSE 0
        END AS win_flag
    FROM fact_match_result f
    JOIN dim_date d
        ON f.date_key = d.date_key
    JOIN dim_team at
        ON f.away_team_key = at.team_key
)

SELECT
    year,
    team,
    COUNT(*) AS matches_played,
    SUM(win_flag) AS wins,
    ROUND(SUM(win_flag) * 1.0 / COUNT(*), 3) AS win_rate
FROM team_match_results
GROUP BY year, team
HAVING COUNT(*) > 0
ORDER BY team, year;
```

Wins by team and tournament

```

WITH team_tournament_wins AS (
    SELECT
        t.tournament_name,
        ht.team_name AS team,
        CASE
            WHEN f.home_win = 1 THEN 1
            ELSE 0
        END AS win_flag
    FROM fact_match_result f
    JOIN dim_tournament t
        ON f.tournament_key = t.tournament_key
    JOIN dim_team ht
        ON f.home_team_key = ht.team_key

    UNION ALL

    SELECT
        t.tournament_name,
        at.team_name AS team,
        CASE
            WHEN f.away_win = 1 THEN 1
            ELSE 0
        END AS win_flag
    FROM fact_match_result f
    JOIN dim_tournament t
        ON f.tournament_key = t.tournament_key
    JOIN dim_team at
        ON f.away_team_key = at.team_key
)

SELECT
    tournament_name,
    team,
    SUM(win_flag) AS wins
FROM team_tournament_wins
GROUP BY tournament_name, team
HAVING SUM(win_flag) > 0
ORDER BY tournament_name, wins DESC, team;

```

Counts of Home and Away Wins for each team

```
WITH home_away_wins AS (  
    SELECT  
        ht.team_name AS team,  
        'Home' AS venue_type,  
        COUNT(*) AS wins  
    FROM fact_match_result f  
    JOIN dim_team ht  
        ON f.home_team_key = ht.team_key  
    WHERE f.home_win = 1  
    GROUP BY ht.team_name  
  
    UNION ALL  
  
    SELECT  
        at.team_name AS team,  
        'Away' AS venue_type,  
        COUNT(*) AS wins  
    FROM fact_match_result f  
    JOIN dim_team at  
        ON f.away_team_key = at.team_key  
    WHERE f.away_win = 1  
    GROUP BY at.team_name  
)  
  
SELECT  
    team,  
    venue_type,  
    wins  
FROM home_away_wins  
ORDER BY team, venue_type;
```

Grouping matches by goal difference and outcome

```
SELECT
    FLOOR(goal_diff / 2.0) * 2 AS goal_diff_bin,
    CASE
        WHEN draw = 1 THEN 'Draw'
        WHEN home_win = 1 THEN 'Home Win'
        WHEN away_win = 1 THEN 'Away Win'
    END AS match_outcome,
    COUNT(*) AS match_count
FROM fact_match_result
GROUP BY
    FLOOR(goal_diff / 2.0) * 2,
    CASE
        WHEN draw = 1 THEN 'Draw'
        WHEN home_win = 1 THEN 'Home Win'
        WHEN away_win = 1 THEN 'Away Win'
    END
ORDER BY goal_diff_bin, match_outcome;
```